Unit 2 Progress Check: MCQ

1. On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $Mg(NO_3)_2(s)$?

(A) There is polar covalent bonding between Mg atoms and N atoms.

(B) There is polar covalent bonding between Mg atoms and O atoms.

(C) There is ionic bonding between N^{5+} ions and O^{2-} ions.

(D) There is ionic bonding between Mg^{2+} ions and NO_3^- ions. ✓

Answer D

Correct. There is ionic bonding in solid $Mg(NO_3)_2$ due to coulombic attractions between Mg^{2+} ions and NO_3^- ions. (There is some polar covalent bonding between N and O within the NO_3^- ion.)

2. Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

Answer D

Correct. A relatively large difference between the electronegativities of the two elements indicates that one element has a tendency to form cations and the other has a tendency to form anions.

3. Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?

(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above $1000^\circ C$.

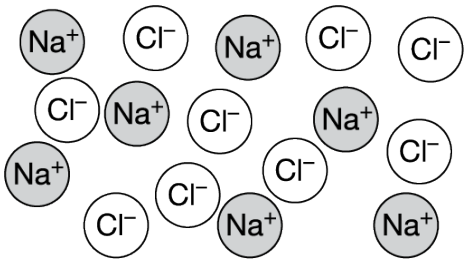
(D) The compound contains strong ionic bonds.

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Answer B

Correct. Elements with the same electronegativity are more likely to share electrons and form a nonpolar covalent bond.

4.



Molten (liquid) NaCl is represented by the particulate diagram shown above. Which of the following indicates whether $\mathrm{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because Na is a metal.

(B) It conducts electricity because ions are free to move. ✓

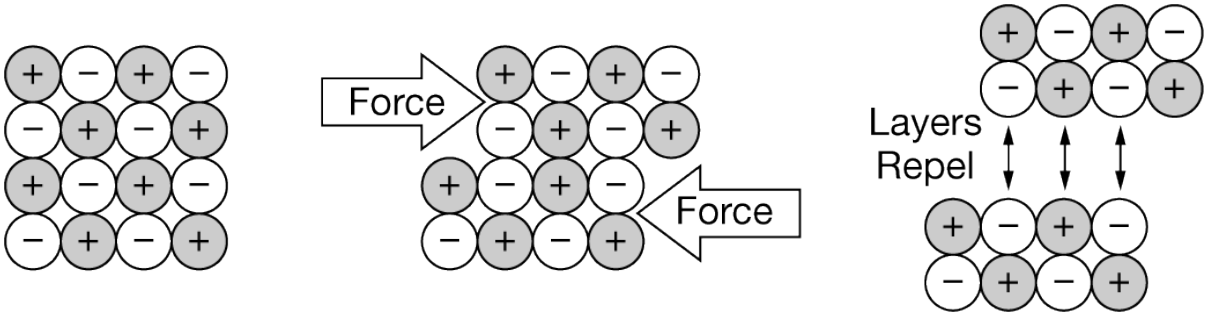
(C) It does not conduct electricity because Cl is a nonmetal.

(D) It does not conduct electricity because there are no free electrons.

Answer B

Correct. For a substance to conduct electricity, it must contain charged particles that are mobile. In molten salts such as NaCl, those charged particles are ions. (Metals conduct electricity via mobile electrons within a lattice of metal atoms.)

5.



The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

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(A) Density

(B) Brittleness ✓

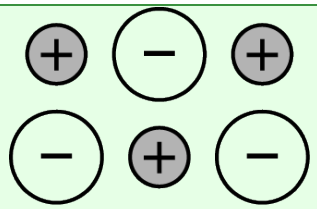
(C) Malleability

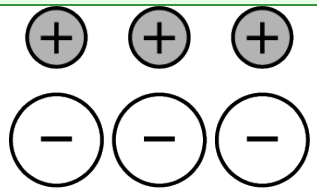
(D) Conductivity

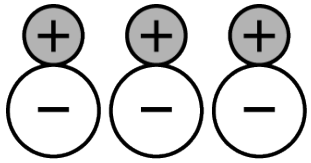
Answer B

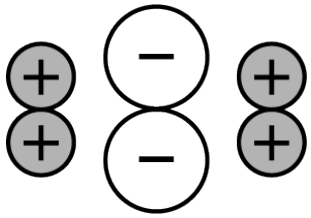
Correct. The brittleness of ionic solids is due to the internal repulsion of ion layers within the crystal lattice that occurs when the layers are displaced by an outside force parallel to the layers and they move relative to adjacent layers.

6. The particles in solid KI, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid KI?

(A)  ✓

(B) 

(C) 

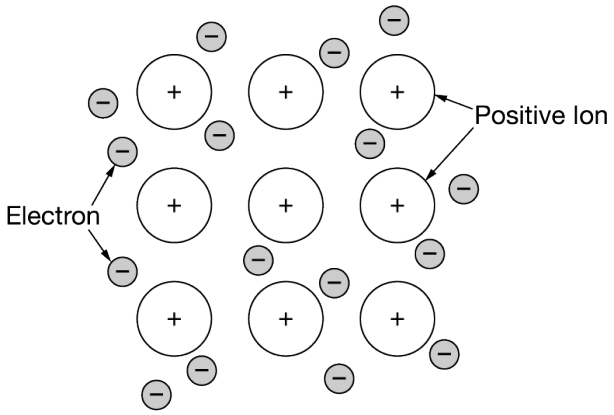
(D) 

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Answer A

Correct. This diagram shows both types of charged ions being surrounded by ions of the opposite charge, thus maximizing attraction and minimizing repulsion.

7.



Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative particles.

(B) It conducts electricity because electrons are free to move through the substance. ✓

(C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.

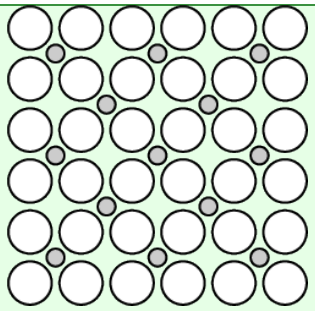
(D) It does not conduct electricity because the positive particles are not free to move through the substance.

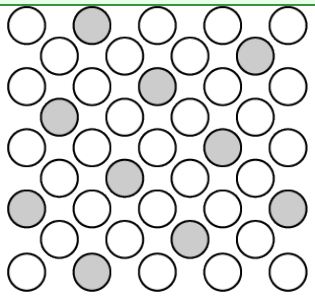
Answer B

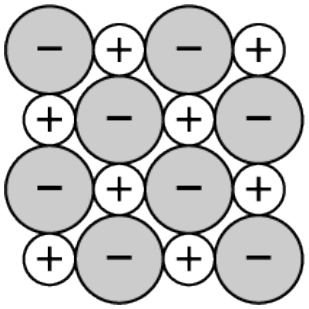
Correct. The diagram shows the structure of a metallic solid, which consists of a lattice of positive ions immersed in a sea of electrons in which electrons can move freely.

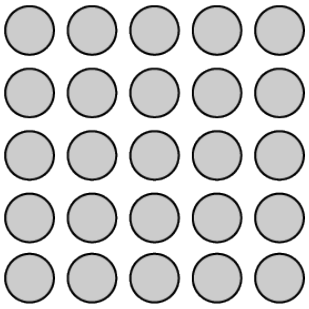
8. Which of the following particulate-level diagrams best represents an interstitial alloy?

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(A) 

(B) 

(C) 

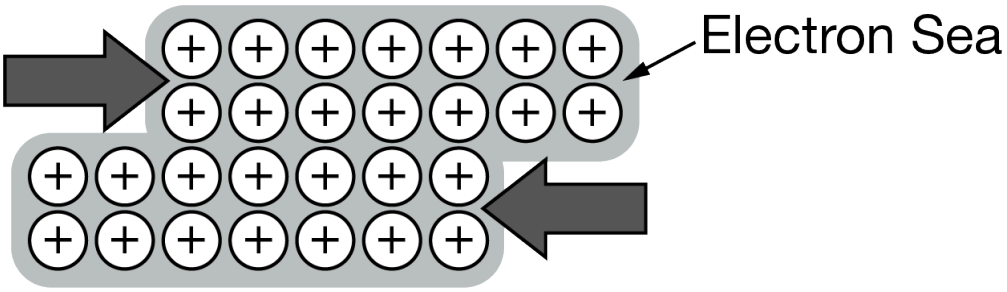
(D) 

Answer A

Correct. In an interstitial alloy, atoms of one of the constituents are significantly smaller than the atoms of the other constituent and occupy the smaller spaces (interstices) among the larger atoms.

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9.



The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

(A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.

(B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓

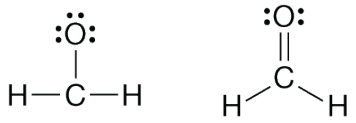
(C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.

(D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.

Answer B

Correct. The diagram shows the structure of a metallic solid, which has adjacent layers of positive ions sliding past one another (as indicated by the arrows) while staying within the shading of the electron sea.

10.



Which of the diagrams above best represents the CH_2O molecule, and why?

(A) Diagram 1, because all bond angles are 180°.

(B) Diagram 1, because all atoms have a formal charge of 0.

(C) Diagram 2, because the molecule has a trigonal pyramidal shape.

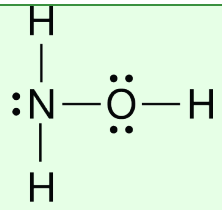
(D) Diagram 2, because all atoms have a formal charge of 0. ✓

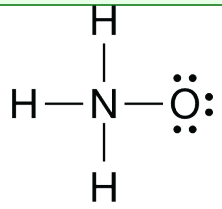
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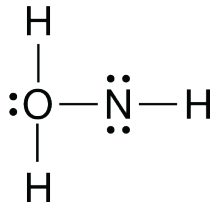
Answer D

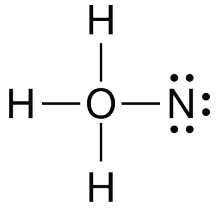
Correct. Diagram 2 represents a molecule in which both C and O have a complete octet and all atoms have a formal charge of zero.

11. Based on formal charges, which of the following is the best Lewis electron-dot diagram for H_3NO?

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. When the formal charge of each atom in this diagram is determined using the rule (formal charge of an atom = the number of valence electrons of the atom − (the number of electrons in lone pairs + half the number of electrons in bonds with other atoms)), the sum of the formal charges on all the atoms in H_3NO is equal to zero. A sum of formal charges on all atoms equal to zero is preferable.

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12.

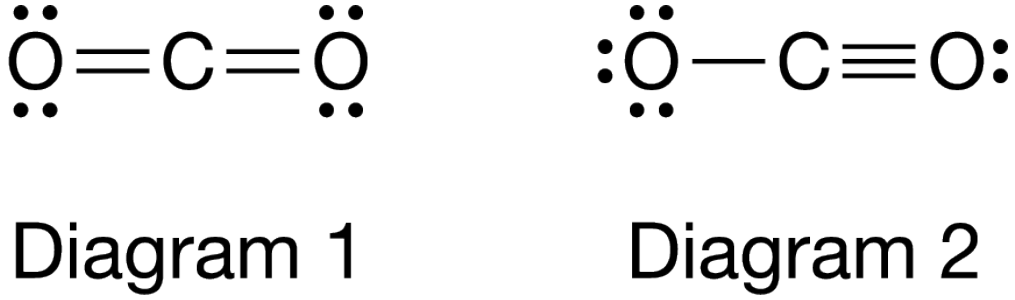


Diagram 1 Diagram 2

Which of the Lewis diagrams shown above is the more likely structure of CO_2, and why?

(A) Diagram 1, because all the atoms have a formal charge of 0. ✓

(B) Diagram 1, because double bonds are stronger than triple bonds.

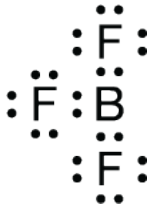
(C) Diagram 2, because all the atoms have a formal charge of 0.

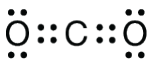
(D) Diagram 2, because triple bonds are stronger than double bonds.

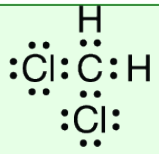
Answer A

Correct. When the formal charge of each atom in diagram 1 is determined using the rule $\left(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - \left(\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}\right)\right)$, the results are zero for each of the three atoms. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero. And in diagram 2, the results for the three atoms (going left to right) are -1, 0, and $+1$, respectively.

13. Which of the following Lewis diagrams represents a molecule that is polar?

(A) 

(B) 

(C)  ✓

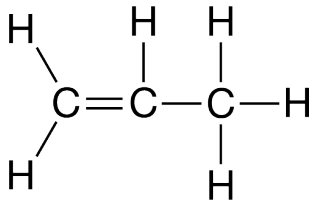
(D) 

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Answer C

Correct. In the CH_2Cl_2 molecule, the H and Cl atoms are in a tetrahedral arrangement around the central C atom. The two electronegative chlorine atoms create a partial negative charge on their side of the carbon atom, leaving a partial positive charge on the opposite (hydrogen) side of the carbon atom. This separation of charge constitutes a dipole, and the molecule is polar.

14.



Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?

(A) The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓

(B) The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is 180°.

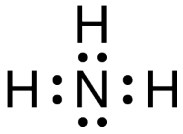
(C) The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.

(D) The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

Answer A

Correct. Double bonds between two atoms are always stronger than single bonds between those same two atoms because in a double bond there are more electrons between the two positive nuclei of the atoms, which results in a stronger bond.

15.



Based on the Lewis diagram for NH_3, shown above, the H-N-H bond angle is closest to which of the following?

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(A) 60°

(B) 90°

(C) 109.5° ✓

(D) 120°